

TRIGONOMETRY

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1. If $A + B = C$, then $\tan A \tan B \tan C = ?$

यदि $A + B = C$ है, तो $\tan A \tan B \tan C$ का मान क्या होगा?

- a) $\tan C + \tan A - \tan B$
 b) $\tan C + \tan A + \tan B$
 c) $\tan A - \tan B - \tan C$
 d) $\tan C - \tan A - \tan B$

2. If $\tan A - \tan B - \tan C = \tan A \tan B \tan C$, what is the value of A in terms of B and C ?

यदि $\tan A - \tan B - \tan C = \tan A \tan B \tan C$ है, तो B और C के संदर्भ में A का मान क्या होगा?

- a) $A = B + C$ b) $A = 2B - 2C$
 c) $A = B - C$ d) $A = \frac{B-C}{2}$

3. Find the value of $\tan 27^\circ \tan 34^\circ + \tan 34^\circ \tan 29^\circ + \tan 29^\circ \tan 27^\circ$.
 $\tan 27^\circ \tan 34^\circ + \tan 34^\circ \tan 29^\circ + \tan 29^\circ \tan 27^\circ$ का मान ज्ञात कीजिए।

- a) 0 b) -1
 c) $\sqrt{3}$ d) 1

4. $\tan 3A - \tan 2A - \tan A =$

- a) $\tan 3A \tan 2A \tan A$
 b) $-\tan 3A \tan 2A \tan A$
 c) $\tan A \tan 2A - \tan 2A \tan 3A - \tan 3A \tan A$
 d) None of these

5. If in $\triangle ABC$, $\tan A + \tan B + \tan C = 6$, then $\cot A \cot B \cot C =$

अगर $\triangle ABC$ में, $\tan A + \tan B + \tan C = 6$ है, तो $\cot A \cot B \cot C =$

- a) 6 b) 1 c) $\frac{1}{6}$ d) None of these

6. $\tan 20^\circ + \tan 40^\circ + \sqrt{3} \tan 20^\circ \tan 40^\circ$ is equal to

$\tan 20^\circ + \tan 40^\circ + \sqrt{3} \tan 20^\circ \tan 40^\circ$ बराबर है

- a) $\frac{\sqrt{3}}{4}$ b) $\frac{\sqrt{3}}{2}$ c) $\sqrt{3}$ d) 1

7. The Value of $(1 + \tan 10^\circ)(1 + \tan 35^\circ)$ is :

$(1 + \tan 10^\circ)(1 + \tan 35^\circ)$ का मान क्या होगा?

- a) $\frac{3}{4}$ b) 1 c) 2 d) $\frac{1}{2}$

8. What is the value of the following in terms of trigonometric ratios?

$$\frac{\sin A}{1 + \cos A} + \frac{1 + \cos A}{\sin A}$$

त्रिकोणमितीय अनुपातों के संदर्भ में निम्नलिखित का मान क्या होगा?

$$\frac{\sin A}{1 + \cos A} + \frac{1 + \cos A}{\sin A}$$

- a) $2 \operatorname{cosec} A$ b) $2 \cos A$
 c) $2 \sec A$ d) $2 \sin A$

9. $(\operatorname{cosec} \theta - \cot \theta)^2 = ?$, $(0^\circ < \theta < 90^\circ)$

- a) $\frac{1 + \cos \theta}{1 - \sin \theta}$ b) $\frac{1 + \cos \theta}{1 - \cos \theta}$
 c) $\frac{1 - \sin \theta}{1 + \cos \theta}$ d) $\frac{1 - \cos \theta}{1 + \cos \theta}$

10. If $\frac{\cos \theta}{1 - \sin \theta} + \frac{\cos \theta}{1 + \sin \theta} = 4$, $0^\circ < \theta < 90^\circ$, then the value of $(\tan \theta + \operatorname{cosec} \theta)$ is:

यदि $\frac{\cos \theta}{1 - \sin \theta} + \frac{\cos \theta}{1 + \sin \theta} = 4$, $0^\circ < \theta < 90^\circ$, तो $(\tan \theta + \operatorname{cosec} \theta)$ का मान है:

- a) $\frac{5\sqrt{2}}{2}$ b) $\frac{5\sqrt{3}}{3}$
 c) $\frac{4\sqrt{3}}{3}$ d) $\frac{5\sqrt{2}}{3}$

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11. The value of $\frac{\sin A}{\cot A + \operatorname{cosec} A} - \frac{\sin A}{\cot A - \operatorname{cosec} A} + 1$ is:

$$\frac{\sin A}{\cot A + \operatorname{cosec} A} - \frac{\sin A}{\cot A - \operatorname{cosec} A} + 1 \text{ का मान है-}$$

- a) $\frac{1}{2}$ b) 3 c) 0 d) 2

a) $\cos \theta$

c) $\cot \theta$

b) $\sin \theta$

d) $\operatorname{cosec} \theta$

12. If $\left\{ \left(\frac{\sec \theta - 1}{\sec \theta + 1} \right)^n = \operatorname{cosec} \theta - \cot \theta \right\}$, then $n = ?$

यदि $\left\{ \left(\frac{\sec \theta - 1}{\sec \theta + 1} \right)^n = \operatorname{cosec} \theta - \cot \theta \right\}$ है, तो $n = ?$

- a) 1 b) 0.5
c) -1 d) -0.5

17. If $\sqrt{\frac{1-\cos \theta}{1+\cos \theta}} \times \sqrt{\frac{\operatorname{cosec} \theta - \cot \theta}{\operatorname{cosec} \theta + \cot \theta}} = \frac{1-r}{1+r}$ then the value of r is:

यदि $\sqrt{\frac{1-\cos \theta}{1+\cos \theta}} \times \sqrt{\frac{\operatorname{cosec} \theta - \cot \theta}{\operatorname{cosec} \theta + \cot \theta}} = \frac{1-r}{1+r}$ है, तो r का मान है:

- a) $\sin \theta$ b) $\operatorname{Cosec} \theta$
c) $\sec \theta$ d) $\cos \theta$

13. $\frac{1+\sin \theta}{\cos \theta}$ is equal to which of the following

(where $\theta \neq \frac{\pi}{2}$)?

$\frac{1+\sin \theta}{\cos \theta}$ निम्नलिखित में से किसके बराबर है?

(जहाँ $\theta \neq \frac{\pi}{2}$)?

- a) $\frac{1+\cos \theta}{\sin \theta}$ b) $\frac{\tan \theta + 1}{\tan \theta - 1}$
c) $\frac{\tan \theta - 1}{\tan \theta + 1}$ d) $\frac{1}{1 - \sin \theta}$

18. $\frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1} = ?$

- a) $\sec \theta \sin \theta$ b) $\sec \theta \tan \theta$
c) $\sec \theta + \tan \theta$ d) $\sec \theta - \tan \theta$

19. The value of $\frac{\sin \theta + \cos \theta - 1}{\sin \theta - \cos \theta + 1} \times \sqrt{\frac{1+\sin \theta}{1-\sin \theta}}$

$\frac{\sin \theta + \cos \theta - 1}{\sin \theta - \cos \theta + 1} \times \sqrt{\frac{1+\sin \theta}{1-\sin \theta}}$ का मान

- a) 1 b) -1
c) -2 d) 2

14. The value of $1 + \sqrt{\frac{\cot \theta + \cos \theta}{\cot \theta - \cos \theta}}$, if $0^\circ < \theta < 90^\circ$, is equal to:

यदि $0^\circ < \theta < 90^\circ$ है, तो $1 + \sqrt{\frac{\cot \theta + \cos \theta}{\cot \theta - \cos \theta}}$ का

मान ज्ञात कीजिए।

- a) $1 - \sec \theta + \tan \theta$ b) $1 - \sec \theta - \tan \theta$
c) $1 + \sec \theta - \tan \theta$ d) $1 + \sec \theta + \tan \theta$

20. Find the value $\frac{\sin \theta + \cos \theta - 1}{\sin \theta - \cos \theta + 1} \times \frac{\tan^2 \theta (\operatorname{cosec}^2 \theta - 1)}{\sec \theta - \tan \theta}$

मान ज्ञात कीजिए, $\frac{\sec \theta - \tan \theta}{\sin \theta + \cos \theta - 1} \times$

$\frac{\tan^2 \theta (\operatorname{cosec}^2 \theta - 1)}{\sec \theta - \tan \theta}$

- a) 0 b) -1
c) 1 d) $\frac{1}{2}$

15. The value of $2 - \sqrt{\frac{\cot \theta + \cos \theta}{\cot \theta - \cos \theta}}$, when $0^\circ < \theta < 90^\circ$ is equal to :

$2 - \sqrt{\frac{\cot \theta + \cos \theta}{\cot \theta - \cos \theta}}$ का मान बताइए, जबकि $0^\circ < \theta < 90^\circ$ है।

- a) $2 + \sec \theta + \tan \theta$ b) $2 - \sec \theta + \tan \theta$
c) $2 - \sec \theta - \tan \theta$ d) $2 + \sec \theta + \tan \theta$

16. If $\operatorname{cosec} \theta + \cot \theta = p$, then the value of $\frac{p^2 - 1}{p^2 + 1}$ is:

यदि $\operatorname{cosec} \theta + \cot \theta = p$, तो $\frac{p^2 - 1}{p^2 + 1}$ का मान है।

21. $\frac{(1+\cos \theta)(\operatorname{Cosec} \theta - \cot \theta) \sec \theta}{\sin \theta (1 - \sin \theta)(\sec \theta + \tan \theta)} =$

- a) $\sec^2 \theta$ b) $\sin^2 \theta$
c) $\cos^2 \theta$ d) $\operatorname{cosec}^2 \theta$

22. For $0 < \theta < 90$, if $\frac{\sec \theta (1 - \sin \theta)(\sec \theta + \tan \theta)}{(\sec \theta - \tan \theta)^2} =$

$\frac{1+k}{1-k}$, then k is equal to:



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यदि $0 < \theta < 90^\circ$ है और $\frac{\sec\theta(1-\sin\theta)(\sec\theta+\tan\theta)}{(\sec\theta-\tan\theta)^2} = \frac{1+k}{1-k}$ है, तो k बराबर है:

- a) $\operatorname{cosec} \theta$ b) $\cos \theta$
c) $\sec \theta$ d) $\sin \theta$

23. $1 + 2 \tan^2 \theta + 2 \sin \theta \sec^2 \theta$, $0^\circ < \theta < 90^\circ$, is equal to:
 $1 + 2 \tan^2 \theta + 2 \sin \theta \sec^2 \theta$, $0^\circ < \theta < 90^\circ$ का मान ज्ञात करें।

- a) $\frac{1-\sin \theta}{1+\sin \theta}$ b) $\frac{1+\sin \theta}{1-\sin \theta}$
c) $\frac{1-\cos \theta}{1+\cos \theta}$ d) $\frac{1+\cos \theta}{1-\cos \theta}$

24. If $1 + 2 \tan^2 \theta + 2 \sin \theta \sec^2 \theta = \frac{a}{b}$, $0^\circ < \theta < 90^\circ$, then $\frac{a+b}{a-b} = ?$

यदि $1 + 2 \tan^2 \theta + 2 \sin \theta \sec^2 \theta = \frac{a}{b}$, $0^\circ < \theta < 90^\circ$ है, तो $\frac{a+b}{a-b}$ क्या होगा?
a) $\operatorname{cosec} \theta$ b) $\cos \theta$
c) $\sec \theta$ d) $\sin \theta$

25. If $\sec\theta + \tan\theta = k$, $0^\circ < \theta < 90^\circ$, then $\frac{(k-1)^2-2}{(k+1)^2-2k}$ is equal to:

यदि $\sec\theta + \tan\theta = k$, $0^\circ < \theta < 90^\circ$ तो $\frac{(k-1)^2-2}{(k+1)^2-2k}$ इसके बराबर है:
a) $\sin\theta - \cos\theta$ b) $\cos\theta + \sin\theta$
c) $\sec\theta \operatorname{cosec}\theta$ d) $\cos\theta \sin\theta$

26. If $x = a \cos \theta + b \sin \theta$ and $y = a \sin \theta - b \cos \theta$, the value of $x^2 + y^2$ is:

यदि $x = a \cos \theta + b \sin \theta$ और $y = a \sin \theta - b \cos \theta$ है, तो $x^2 + y^2$ का मान है:
a) $a^2 - b^2$ b) $a - b$
c) $a^2 + b^2$ d) $a + b$

27. The elimination of θ from $x \cos \theta - y \sin \theta = 2$ and $x \sin \theta - y \cos \theta = 4$ will give:

$x \cos \theta - y \sin \theta = 2$ और $x \sin \theta - y \cos \theta = 4$ से θ का विलोपन देगा:
a) $3x^2 - y^2 = 20$

- b) $x^2 + y^2 = 20$
c) $3x^2 + y^2 = 20$
d) $x^2 - y^2 = 20$

28. If $3 \sin x + 4 \cos x = 2$, then the value of $3 \cos x - 4 \sin x$ is equal to:

यदि $3 \sin x + 4 \cos x = 2$, तो $3 \cos x - 4 \sin x$ का मान है:

- a) $\sqrt{21}$ b) $\sqrt{23}$
c) 21 d) $\sqrt{29}$

29. If $\sin\theta - \cos\theta = \frac{4}{5}$, then find the value of $\sin\theta + \cos\theta$.

यदि $\sin\theta - \cos\theta = \frac{4}{5}$ है, तो $\sin\theta + \cos\theta$ का मान ज्ञात कीजिए।

- a) $\frac{5}{\sqrt{34}}$ b) $\frac{5}{\sqrt{24}}$
c) $\frac{\sqrt{34}}{5}$ d) $\frac{\sqrt{24}}{5}$

30. If θ is an acute angle, and it is given that $5 \sin \theta + 12 \cos \theta = 13$, then what is the value of $\tan \theta$?

यदि θ एक न्यून कोण है, और यह दिया गया है कि $5 \sin \theta + 12 \cos \theta = 13$, तो $\tan \theta$ का मान क्या है?

- a) $\frac{5}{13}$ b) $\frac{13}{12}$
c) $\frac{12}{13}$ d) $\frac{5}{12}$

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31. The Value of $\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta} \times \sqrt{\tan^2 \theta - \sin^2 \theta}$ is equal to:

$\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta} \times \sqrt{\tan^2 \theta - \sin^2 \theta}$ का मान इसके बराबर है:

- a) $\operatorname{cosec} \theta \sec^2 \theta$



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- b) $\sin \theta \sec^2 \theta$
 c) $\sin \theta \cos^2$
 d) $\operatorname{cosec} \theta \cos^2 \theta$

32. If $\tan^2 \theta = 1 - a^2$, then the value of $\sec \theta + \tan^3 \theta \operatorname{cosec} \theta$ is:

यदि $\tan^2 \theta = 1 - a^2$ है, तो $\sec \theta + \tan^3 \theta \operatorname{cosec} \theta$ का मान क्या है?

- a) $(2 - a^2)^{\frac{3}{2}}$ b) $(a^2 - 1)^{\frac{3}{2}}$
 c) $(2 - a^2)^{\frac{3}{2}}$ d) $a^{\frac{3}{2}}$

33. If $\tan^2 \alpha = 3 + Q^2$, then $\sec \alpha + \tan^3 \alpha \operatorname{cosec} \alpha = ?$
 $\tan^2 \alpha = 3 + Q^2$ है, तो $\sec \alpha + \tan^3 \alpha \operatorname{cosec} \alpha = ?$

- a) $(3 + Q^2)^{\frac{3}{2}}$ b) $(7 + Q^2)^{\frac{3}{2}}$
 c) $(5 - Q^2)^{\frac{3}{2}}$ d) $(4 + Q^2)^{\frac{3}{2}}$

34. The expression $\frac{(1 - 2 \sin^2 \theta \cos^2 \theta)(\cot \theta + 1) \cos \theta}{(\sin^4 \theta + \cos^4 \theta)(1 + \tan \theta) \operatorname{cosec} \theta} - 1$, $0^\circ < \theta < 90^\circ$, equals:

व्यंजक $\frac{(1 - 2 \sin^2 \theta \cos^2 \theta)(\cot \theta + 1) \cos \theta}{(\sin^4 \theta + \cos^4 \theta)(1 + \tan \theta) \operatorname{cosec} \theta} - 1$ का

मान बताइए, जहाँ $0^\circ < \theta < 90^\circ$ है।

- a) $\cos^2 \theta$ b) $-\sin^2 \theta$
 c) $\sec^2 \theta$ d) $-\sec^2 \theta$

35. $\frac{(1 + \sec \theta \operatorname{cosec} \theta)^2 (\sec \theta - \tan \theta)^2 (1 + \sin \theta)}{(\sin \theta + \sec \theta)^2 + (\cos \theta + \operatorname{cosec} \theta)^2}$, $0^\circ < \theta < 90^\circ$, is equal to:

$\frac{(1 + \sec \theta \operatorname{cosec} \theta)^2 (\sec \theta - \tan \theta)^2 (1 + \sin \theta)}{(\sin \theta + \sec \theta)^2 + (\cos \theta + \operatorname{cosec} \theta)^2}$, $0^\circ < \theta < 90^\circ$, का मान इनमें से किसके बराबर होगा?

- a) $1 - \sin \theta$ b) $\sin \theta$
 c) $1 - \cos \theta$ d) $\cos \theta$

36. $\left(\frac{\tan^3 \theta}{\sec^2 \theta} + \frac{\cot^3 \theta}{\operatorname{cosec}^2 \theta} + 2 \sin \theta \cos \theta \right) \div (1 + \operatorname{cosec}^2 \theta + \tan^2 \theta)$, $0^\circ < \theta < 90^\circ$, is equal to:

$$\left(\frac{\tan^3 \theta}{\sec^2 \theta} + \frac{\cot^3 \theta}{\operatorname{cosec}^2 \theta} + 2 \sin \theta \cos \theta \right) \div$$

$(1 + \operatorname{cosec}^2 \theta + \tan^2 \theta)$, $0^\circ < \theta < 90^\circ$, का मान इनमें से किसके बराबर होगा?

- a) $\sin \theta \cos \theta$ b) $\sec \theta$
 c) $\operatorname{cosec} \theta \sec \theta$ d) $\operatorname{cosec} \theta$

37. $\frac{1 + \cos \theta - \sin^2 \theta}{\sin \theta (1 + \cos \theta)} \times \frac{\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta}}{\tan \theta + \cot \theta}$, $0^\circ < \theta < 90^\circ$, is equal to:

$$\frac{1 + \cos \theta - \sin^2 \theta}{\sin \theta (1 + \cos \theta)} \times \frac{\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta}}{\tan \theta + \cot \theta}, 0^\circ < \theta < 90^\circ$$

का मान इनमें से किसके बराबर होगा?

- a) $\tan \theta$ b) $\operatorname{cosec} \theta$
 c) $\sec \theta$ d) $\cot \theta$

38. The expression $\frac{\tan^6 \theta - \sec^6 \theta + 3 \sec^2 \theta \tan^2 \theta}{\tan^2 \theta + \cot^2 \theta + 2}$, $0^\circ < \theta < 90^\circ$, is equal to:

व्यंजक $\frac{\tan^6 \theta - \sec^6 \theta + 3 \sec^2 \theta \tan^2 \theta}{\tan^2 \theta + \cot^2 \theta + 2}$ का मान

बताइए जहाँ $0^\circ < \theta < 90^\circ$ है।

- a) $\sec^2 \theta \operatorname{cosec}^2 \theta$ b) $-\sec^2 \theta \operatorname{cosec}^2 \theta$
 c) $\cos^2 \theta \sin^2 \theta$ d) $-\cos^2 \theta \sin^2 \theta$

39. If $0^\circ < \theta < 90^\circ$, then $\frac{(1 - \sin \theta)(\sec \theta + \tan \theta) \tan \theta}{(\tan \theta + \sec \theta + 1)(\cot \theta - \operatorname{cosec} \theta + 1)} = ?$

यदि $0^\circ < \theta < 90^\circ$ है, तो $\frac{(1 - \sin \theta)(\sec \theta + \tan \theta) \tan \theta}{(\tan \theta + \sec \theta + 1)(\cot \theta - \operatorname{cosec} \theta + 1)}$ का मान ज्ञात करें।

- a) $\frac{\sin \theta}{2}$ b) $\sin \theta$
 c) $\cos \theta$ d) $\frac{\cos \theta}{2}$

40. The expression $\frac{\cos^4 \theta - \sin^4 \theta + 2 \sin^2 \theta + 3}{(\operatorname{cosec} \theta + \cot \theta + 1)(\operatorname{cosec} \theta - \cot \theta + 1) - 2}$, $0^\circ < \theta < 90^\circ$, is equal to:

$$\frac{\cos^4 \theta - \sin^4 \theta + 2 \sin^2 \theta + 3}{(\operatorname{cosec} \theta + \cot \theta + 1)(\operatorname{cosec} \theta - \cot \theta + 1) - 2}, 0^\circ < \theta < 90^\circ$$

का मान इनमें से किसके बराबर होगा?

व्यंजक $\frac{\cos^4 \theta - \sin^4 \theta + 2 \sin^2 \theta + 3}{(\operatorname{cosec} \theta + \cot \theta + 1)(\operatorname{cosec} \theta - \cot \theta + 1) - 2}$ का

मान बताइए, जहाँ $0^\circ < \theta < 90^\circ$ है।

- a) $\frac{1}{2} \sin \theta$ b) $\sec \theta$
 c) $2 \operatorname{cosec} \theta$ d) $2 \sin \theta$



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41. If $\tan\theta + \cot\theta = 4$, then the ratio of $3(\tan^2\theta + \cot^2\theta)$ to $(2\operatorname{cosec}^2\theta \sec^2\theta - 4)$ will be:

यदि $\tan\theta + \cot\theta = 4$ है, तो $3(\tan^2\theta + \cot^2\theta)$ का $(2\operatorname{cosec}^2\theta \sec^2\theta - 4)$ से

अनुपात ज्ञात करें।

- a) 4 : 3 b) 3 : 4
c) 5 : 4 d) 3 : 2

42. $\frac{(\sin\theta - \cos\theta)(1 + \tan\theta + \cot\theta)(\operatorname{cosec}^2\theta)}{\sec^3\theta - \operatorname{cosec}^3\theta}$, $0^\circ < \theta < 90^\circ$, is equal to:

$$\frac{(\sin\theta - \cos\theta)(1 + \tan\theta + \cot\theta)(\operatorname{cosec}^2\theta)}{\sec^3\theta - \operatorname{cosec}^3\theta}, 0^\circ < \theta < 90^\circ$$

किसके बराबर होगा?

- a) $\sec\theta \operatorname{cosec}\theta$ b) $\cos^2\theta$
c) $\sin^2\theta$ d) $\sin\theta \cos\theta$

43. If $\cot^2\theta + \cot^4\theta = 2$, then the value of $2\sin^4\theta + \sin^2\theta$ is:

यदि $\cot^2\theta + \cot^4\theta = 2$ है, तो $2\sin^4\theta + \sin^2\theta$ का मान ज्ञात करें।

- a) 3 b) 5 c) 1 d) 2

44. If $\tan^2\theta + \tan^4\theta = 1$, then:

यदि $\tan^2\theta + \tan^4\theta = 1$ है तो _____ होगा।

- a) $\cot^2\theta + \cot^4\theta = 1$
b) $\cos^2\theta + \cos^4\theta = 1$
c) $\sin^2\theta + \sin^4\theta = 1$
d) $\operatorname{cosec}^2\theta + \sec^4\theta = 1$

45. $\frac{\sec^2\theta(2 + \tan^2\theta + \cot^2\theta) \div (\sin^2\theta - \tan^2\theta)}{(\operatorname{cosec}^2\theta + \sec^2\theta)(1 + \cot^2\theta)^2} =$

- a) -1 b) 1
c) -2 d) 2

46. $\frac{\cos^6\theta + \sin^6\theta + 3\sin^2\theta \cos^2\theta}{\operatorname{cosec}\theta \sec\theta (\sin\theta + \cos\theta - 1)(\sin\theta + \cos\theta + 1)} =$

- a) 1 b) 2
c) $\frac{1}{2}$ d) 3

47. $\frac{(1 + \cos\theta)^2 + \sin^2\theta}{(\operatorname{cosec}^2\theta - 1)\sin^2\theta} = ?$

- a) $\cos\theta (1 + \sin\theta)$
b) $2\cos\theta (1 + \sec\theta)$
c) $\sec\theta (1 + \sin\theta)$
d) $2\sec\theta (1 + \sec\theta)$

48. If $\frac{(\sin\theta - \operatorname{cosec}\theta)(\cos\theta - \sec\theta)}{\tan^2\theta - \sin^2\theta} = r^3$, then $r = ?$

- अगर $\frac{(\sin\theta - \operatorname{cosec}\theta)(\cos\theta - \sec\theta)}{\tan^2\theta - \sin^2\theta} = r^3$ है, तो $r = ?$
a) $\sin\theta \cos\theta$ b) $\tan\theta$
c) $\cot\theta$ d) $\operatorname{Cosec}\theta \sec\theta$

49. Solve the following to find its value in terms of trigonometric ratios. $(\sin A + \cos A)(1 - \sin A \cos A)$

निम्नलिखित को त्रिकोणमितीय अनुपातों के संदर्भ में इसका मान ज्ञात करने के लिए हल करें। $(\sin A + \cos A)(1 - \sin A \cos A)$

- a) $\sin^3 A + \cos^3 A$
b) $\sin^2 A - \cos^2 A$
c) $[\cos A - \sin A][\sin^2 A + \cos^2 A]$
d) $\sin^3 A - \cos^3 A$

50. If $\sin\alpha + \cos\alpha = \tan\frac{\pi}{3}$, then the value of $(\sin^3\alpha + \cos^3\alpha)$ is equal to:

यदि $\sin\alpha + \cos\alpha = \tan\frac{\pi}{3}$ है, तो $(\sin^3\alpha + \cos^3\alpha)$ का मान ज्ञात कीजिए।

- a) $\frac{1}{2}$ b) 1 c) 0 d) $\frac{3}{2}$

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51. If $\sin\theta + \cos\theta = \frac{\sqrt{3}-1}{2\sqrt{2}}$, then what is the value of $\tan\theta + \cot\theta$?

यदि $\sin\theta + \cos\theta = \frac{\sqrt{3}-1}{2\sqrt{2}}$ है, तो $\tan\theta + \cot\theta$ का मान क्या है?

- a) $8(\sqrt{3}-2)$ b) $12(\sqrt{3}-2)$
c) $12(\sqrt{3}+2)$ d) $8(\sqrt{3}+2)$

52. What is the average of the sixty terms given below?

$\cos^2 x, \cos^2 2x, \cos^2 3x, \dots, \cos^2 30x, \sin^2 x, \sin^2 2x, \sin^2 3x, \dots, \sin^2 30x$

नीचे दिए गए साठ पदों का औसत क्या होगा?

- $\cos^2 x, \cos^2 2x, \cos^2 3x, \dots, \cos^2 30x, \sin^2 x, \sin^2 2x, \sin^2 3x, \dots, \sin^2 30x$
a) $\cos^2 x$ b) 0.5
c) 1 d) $\cos^2 x \sin^2 x$

53. The value of $\frac{\sec\theta \csc\theta}{2+\tan^2\theta+\cot^2\theta}$ is equal to:

- $\frac{\sec\theta \csc\theta}{2+\tan^2\theta+\cot^2\theta}$ का मान ज्ञात करें।
a) $\sec\theta \csc\theta$ b) $\sec\theta \sin\theta$
c) $\sin\theta \cos\theta$ d) $\cos\theta \csc\theta$

54. Let $0^\circ < \theta < 90^\circ$. $(1 + \cot^2\theta)(1 + \tan^2\theta) \times (\sin\theta - \csc\theta)(\cos\theta - \sec\theta)$ is equal to:

मान लें कि $0^\circ < \theta < 90^\circ$ है। तो $(1 + \cot^2\theta)(1 + \tan^2\theta) \times (\sin\theta - \csc\theta)(\cos\theta - \sec\theta)$ का मान इनमें से किसके बराबर होगा?

- a) $\sec\theta \csc\theta$ b) $\sin\theta + \cos\theta$
c) $\sec\theta + \csc\theta$ d) $\sin\theta \cos\theta$

55. If $7\sin^2\theta + 4\cos^2\theta = 5$ and θ lies in the first quadrant, then what is the value of $\frac{\sqrt{3}\sec\theta + \tan\theta}{\sqrt{2}\cot\theta - \sqrt{3}\cos\theta}$?

यदि $7\sin^2\theta + 4\cos^2\theta = 5$ और θ पहले चतुर्थांश में स्थित है, तो $\frac{\sqrt{3}\sec\theta + \tan\theta}{\sqrt{2}\cot\theta - \sqrt{3}\cos\theta}$ का मान क्या होगा?

- a) $4\sqrt{2}$ b) $2(1 + \sqrt{2})$
c) $3\sqrt{2}$ d) $2(\sqrt{2} - 1)$

56. If $117\cos^2 A + 129\sin^2 A = 120$ and $170\cos^2 B + 158\sin^2 B = 161$, then the value of $\operatorname{Cosec}^2 A \sec^2 B$ is:

अगर $117\cos^2 A + 129\sin^2 A = 120$ और $170\cos^2 B + 158\sin^2 B = 161$ है, तो

$\operatorname{Cosec}^2 A \sec^2 B$ है:

- a) 1 b) 9
c) 4 d) 16

57. $\frac{\sec^6\theta - \tan^6\theta - 3\sec^2\theta \tan^2\theta + 1}{\cos^4\theta \sin^4\theta + 2\sin^2\theta + 2} =$

- a) $\frac{2}{3}$ b) 1
c) $\frac{3}{4}$ d) $\frac{1}{2}$

58. $\cos^3 60 - \cos^3 240 - \cos^3 360 =$ _____
 $\cos^3 60 - \cos^3 240 - \cos^3 360 =$ _____

होगा।

- a) $-\frac{3}{4}$ b) $-\frac{7}{5}$
c) $-\frac{3}{5}$ d) $-\frac{9}{7}$

59. $\tan(\theta - 14\pi)$ is equal to:

$\tan(\theta - 14\pi)$ निम्न में किसके बराबर है?

- a) $\tan\theta$ b) $-\cot\theta$
c) $\cot\theta$ d) $-\tan\theta$

60. What is the value of $\cos\left(-\frac{17\pi}{3}\right)$?

$\cos\left(-\frac{17\pi}{3}\right)$ का मान क्या होगा?

- a) 1 b) $\frac{\sqrt{3}}{2}$
c) $\frac{1}{2}$ d) 0

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61. If $\tan 40^\circ = \alpha$, then find $\frac{\tan 320^\circ - \tan 310^\circ}{1 + \tan 320^\circ \tan 310^\circ}$



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यदि $\tan 40^\circ = \alpha$ है, तो $\frac{\tan 320^\circ - \tan 310^\circ}{1 + \tan 320^\circ \cdot \tan 310^\circ}$ का मान ज्ञात किजिए।

- a) $\frac{1-\alpha^2}{\alpha}$ b) $\frac{1+\alpha^2}{2\alpha}$
c) $\frac{1-\alpha^2}{2\alpha}$ d) $\frac{1+\alpha^2}{\alpha}$

62. If $\cos x = \frac{-\sqrt{3}}{2}$ and $\pi < x < \frac{3\pi}{2}$, then the value of $4 \cot^2 x - 3 \operatorname{cosec}^2 x$ is:

यदि $\cos x = \frac{-\sqrt{3}}{2}$ और $\pi < x < \frac{3\pi}{2}$ है, तो $4 \cot^2 x - 3 \operatorname{cosec}^2 x$ का मान है:

- a) 8 b) 0
c) 2 d) 1

63. If $\cos x = \frac{-1}{2}$ and $\pi < x < \frac{3\pi}{2}$, then the value of $4 \tan^2 x + 3 \operatorname{cosec}^2 x$ is:

यदि $\cos x = \frac{-1}{2}$ और $\pi < x < \frac{3\pi}{2}$ है, तो $4 \tan^2 x + 3 \operatorname{cosec}^2 x$ है:

- a) 16 b) 8
c) 4 d) 10

64. If $\cos x = \frac{-\sqrt{3}}{2}$ and $\pi < x < \frac{3\pi}{2}$, then the value of $2 \cot^2 x + 3 \operatorname{cosec}^2 x$ is:

यदि $\cos x = \frac{-\sqrt{3}}{2}$ और $\pi < x < \frac{3\pi}{2}$ है, तो $2 \cot^2 x + 3 \operatorname{cosec}^2 x$ है:

- a) 14 b) 16
c) 8 d) 18

65. If A lies between 45° and 540° , and $\sin A = 0.5$, what is the value of $A/3$ in degrees?

यदि A का मान 45° और 540° के मध्य है, और $\sin A = 0.5$ है, तो $A/3$ का मान अंश में कितना होगा?

- a) 170° b) 175°
c) 165° d) 160°

66. Find θ , if $\cos \theta = -\frac{\sqrt{3}}{2}$.

यदि $\cos \theta = -\frac{\sqrt{3}}{2}$, है, तो θ का माना ज्ञात करें।

- a) $\frac{3\pi}{2}$ b) $\frac{2\pi}{3}$ c) $\frac{4\pi}{3}$ d) $\frac{5\pi}{6}$

67. Find $\cos \left(-\frac{7\pi}{2}\right)$.
 $\cos \left(-\frac{7\pi}{2}\right)$ ज्ञात कीजिए।

- a) $\frac{1}{2}$ b) 1 c) -1 d) 0

68. Which among the following increases continuously in the range $0^\circ < \theta < 90^\circ$?

निम्नलिखित में से कौन $0^\circ < \theta < 90^\circ$ की सीमा में लगातार बढ़ता है?

- a) $\cot \theta$ b) $\operatorname{Cosec} \theta$
c) $\tan \theta$ d) $\cos \theta$

69. The value of $\frac{3(\cot^2 47^\circ - \sec^2 43^\circ) - 2(\tan^2 23^\circ - \operatorname{cosec}^2 67^\circ)}{\operatorname{cosec}^2 (68^\circ + \theta) - \tan(\theta + 61^\circ) - \tan^2 (22^\circ - \theta) + \cot(29^\circ - \theta)}$ is:

$\frac{3(\cot^2 47^\circ - \sec^2 43^\circ) - 2(\tan^2 23^\circ - \operatorname{cosec}^2 67^\circ)}{\operatorname{cosec}^2 (68^\circ + \theta) - \tan(\theta + 61^\circ) - \tan^2 (22^\circ - \theta) + \cot(29^\circ - \theta)}$ का मान क्या होगा?

- a) 0 b) -1
c) 1 d) 5

70. The value of $\frac{3(\operatorname{cosec}^2 26^\circ - \tan^2 64^\circ) + (\cot^2 42^\circ - \sec^2 48^\circ)}{\cot(22^\circ - \theta) - \operatorname{cosec}^2 (62^\circ + \theta) - \tan(\theta + 68^\circ) + \tan^2 (28^\circ - \theta)}$ is:

$\frac{3(\operatorname{cosec}^2 26^\circ - \tan^2 64^\circ) + (\cot^2 42^\circ - \sec^2 48^\circ)}{\cot(22^\circ - \theta) - \operatorname{cosec}^2 (62^\circ + \theta) - \tan(\theta + 68^\circ) + \tan^2 (28^\circ - \theta)}$ का मान बताइए।

- a) 3 b) 4 c) -1 d) -2

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71. The value of $\frac{(\cos 9^\circ + \sin 81^\circ)(\sec 9^\circ + \operatorname{cosec} 81^\circ)}{\operatorname{cosec}^2 71^\circ + \cos^2 15^\circ - \tan^2 19^\circ + \cos^2 75^\circ}$ is:

$\frac{(\cos 9^\circ + \sin 81^\circ)(\sec 9^\circ + \operatorname{cosec} 81^\circ)}{\operatorname{cosec}^2 71^\circ + \cos^2 15^\circ - \tan^2 19^\circ + \cos^2 75^\circ}$ का मान बताइए।

- a) 1 b) 4 c) -3 d) 2

72. Which of the following values suits for A to make the equation $\frac{A \tan 62^\circ \sec 28^\circ \cot 38^\circ}{\operatorname{Cosec} 62^\circ \tan 11^\circ} = 1$ true?

यदि $\frac{A \tan 62^\circ \sec 28^\circ \cot 38^\circ}{\operatorname{Cosec} 62^\circ \tan 11^\circ} = 1$ है, तो निम्नलिखित में से कौन सा मान A के लिए समीकरण बनाने के लिए उपयुक्त है ?

a) $\frac{\tan 38^\circ}{\tan 79^\circ \tan 28^\circ}$ b) $\frac{\tan 28^\circ \tan 79^\circ}{\tan 38^\circ}$
 c) $\frac{\tan 28^\circ \tan 38^\circ}{\tan 79^\circ}$ d) $\frac{\tan 38^\circ \tan 79^\circ}{\tan 28^\circ}$

73. If $\sin(x) = \frac{2}{5}$ and x is an acute angle, find the exact values of $\cos(4x) - \cos(2x)$.

यदि $\sin(x) = \frac{2}{5}$ और x एक न्यूनकोण है, तो $\cos(4x) - \cos(2x)$ का सटीक मान ज्ञात कीजिए।

- a) $\frac{625}{32}$ b) $-\frac{472}{625}$
 c) $-\frac{427}{625}$ d) $-\frac{472}{25}$

74. The value of $\frac{\sin 4\theta}{(1 - \cos 4\theta)}$ is:

$\frac{\sin 4\theta}{(1 - \cos 4\theta)}$ का मान क्या होगा?

a) $\tan \theta$ b) $\cot \theta$
 c) $\cot 2\theta$ d) $\tan 2\theta$

75. If $\sin \theta + \cos \theta = \sqrt{2}$, then find the value of $\operatorname{Cosec} \theta + \cot \theta$.

यदि $\sin \theta + \cos \theta = \sqrt{2}$ है, तो $\operatorname{Cosec} \theta + \cot \theta$ का मान ज्ञात कीजिए।

- a) $\sqrt{2} - 1$ b) $\sqrt{2} + 5$
 c) $\sqrt{3} - 1$ d) $\sqrt{2} + 1$

76. What is the value of $(4\sin^3 x - 3\sin x + \sin 3x)$?

$(4\sin^3 x - 3\sin x + \sin 3x)$ का मान क्या होगा?

- a) 0 b) 4 c) 1 d) 3

77. If $x + \frac{1}{x} = 2 \cos \theta$, then $x^3 + \frac{1}{x^3}$?

यदि $x + \frac{1}{x} = 2 \cos \theta$ है, तो $x^3 + \frac{1}{x^3} = ?$

- a) $2 \cos 2\theta$ b) $\cos 3\theta$
 c) $2 \cos 3\theta$ d) $\cos 2\theta$

78. What will be the value of $\sin 10^\circ - 4/3 \sin^3 10^\circ$?

$\sin 10^\circ - 4/3 \sin^3 10^\circ$ का मान क्या होगा?

- a) $\frac{1}{3\sqrt{3}}$ b) $1/6$
 c) $\frac{1}{2\sqrt{3}}$ d) $\frac{\sqrt{3}}{6}$

79. If $2 \frac{\cos^2 x - \sec^2 x}{\tan^2 x} = a + b \cos 2x$, then $a, b = ?$

यदि $2 \frac{\cos^2 x - \sec^2 x}{\tan^2 x} = a + b \cos 2x$ है, तो $a, b = ?$

- a) $\frac{-3}{2}, \frac{-1}{2}$ b) $\frac{3}{2}, \frac{1}{2}$
 c) $-3, -1$ d) $3, 1$

80. If $(\cos q + \sin q) = 31/25$ then what will be the value of $\cos^2 q$?

यदि $(\cos q + \sin q) = 31/25$ है, तो $\cos^2 q$ का मान क्या होगा?

- a) $522/625$ b) $512/625$
 c) $513/625$ d) $527/625$

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81. If $\sin x = \frac{2}{3}$, then find the value of $\cos 3x$.

यदि $\sin x = \frac{2}{3}$, तो $\cos 3x$ का मान ज्ञात कीजिए।

- a) 0.6735 b) -0.8765



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c) -0.5797 d) 0.5678

82. $(\operatorname{cosec} A - \sin A)^2 + (\sec A - \cos A)^2 - (\cot A - \tan A)^2 =$

- a) 2 b) 0
c) 1 d) -1

83. $\frac{\sin \theta [(1 - \tan \theta) \tan \theta + \sec^2 \theta]}{(1 - \sin \theta) \tan \theta (1 + \tan \theta) (\sec \theta + \tan \theta)} =$
a) 1 b) $\operatorname{cosec} \theta \sec \theta$
c) $\sin \theta \cos \theta$ d) -1

84. The expression $\frac{(1 - \sin \theta + \cos \theta)^2 (1 - \cos \theta) \sec^3 \theta \operatorname{cosec}^2 \theta}{(\sec \theta - \tan \theta) (\tan \theta + \cot \theta)}$, $0^\circ < \theta < 90^\circ$

90, is equal to:

व्यंजक $\frac{(1 - \sin \theta + \cos \theta)^2 (1 - \cos \theta) \sec^3 \theta \operatorname{cosec}^2 \theta}{(\sec \theta - \tan \theta) (\tan \theta + \cot \theta)}$ का

मान बताइए, जहाँ $0^\circ < \theta < 90^\circ$ है।

- a) $\sin \theta$ b) $2 \cos \theta$
c) $\cot \theta$ d) $2 \tan \theta$

85. The expression $\frac{(1 + \cos \theta + \sin \theta) \div (1 + \cos \theta - \sin \theta)}{(\sec \theta - \cos \theta) (\cot \theta + \tan \theta) (1 + \sin \theta)}$, $0^\circ < \theta < 90^\circ$, is equal to:

व्यंजक $\frac{(1 + \cos \theta + \sin \theta) \div (1 + \cos \theta - \sin \theta)}{(\sec \theta - \cos \theta) (\cot \theta + \tan \theta) (1 + \sin \theta)}$, जहाँ

$0^\circ < \theta < 90^\circ$ है, इनमें से किसके बराबर होगा?

- a) $\tan \theta$ b) $\cot \theta$
c) $\sec \theta$ d) $\sin \theta$

86. The value of $\frac{\sec \phi (1 - \sin \phi) (\sin \phi + \cos \phi) (\sec \phi + \tan \phi)}{\sin \phi (1 + \tan \phi) + \cos \phi (1 + \cot \phi)}$

is equal to:

$\frac{\sec \phi (1 - \sin \phi) (\sin \phi + \cos \phi) (\sec \phi + \tan \phi)}{\sin \phi (1 + \tan \phi) + \cos \phi (1 + \cot \phi)}$ का मान है:

- a) $2 \cos \phi$ b) $\operatorname{cosec} \phi \sec \phi$
c) $2 \sin \phi$ d) $\sin \phi \cos \phi$

87. If $m = \sec \theta - \tan \theta$ and $n = \operatorname{cosec} \theta + \cot \theta$, then what is the value of $m + n(m - 1)$?

यदि $m = \sec \theta - \tan \theta$ और $n = \operatorname{cosec} \theta + \cot \theta$ है, तो $m + n(m - 1)$ का मान क्या है?

- a) 0 b) 2
c) 1 d) -1

88. If $\sin m + \sin n = p$, $\cos m + \cos n = q$, then find the value of $\sin m \times \sin n + \cos m \times \cos n$.

यदि $\sin m + \sin n = p$, $\cos m + \cos n = q$, तो $\sin m \times \sin n + \cos m \times \cos n$ का मान ज्ञात कीजिए।

- a) $p^2 + q^2 - 2$
b) $(p^2 + q^2 - 2)/2$
c) $p + q - pq$
d) $p + q + pq$

89. $\frac{(2 \sin A)(1 + \sin A)}{1 + \sin A + \cos A} =$
a) $1 + \sin A - \cos A$
b) $1 - \sin A \cos A$
c) $1 + \cos A - \sin A$
d) $1 + \sin A \cos A$

90. What is the value of the expression $\cos 2A \cos 2B + \sin^2(A - B) - \sin^2(A + B)$?

व्यंजक $\cos 2A \cos 2B + \sin^2(A - B) - \sin^2(A + B)$ का मान क्या होगा?

- a) $\sin(2A - 2B)$ b) $\sin(2A + 2B)$
c) $\cos(2A + 2B)$ c) $\cos(2A - 2B)$

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91. If $x = \frac{2 \sin \theta}{(1 + \cos \theta + \sin \theta)}$, then the value of $\frac{1 - \cos \theta + \sin \theta}{1 + \sin \theta}$ is:

यदि $x = \frac{2 \sin \theta}{(1 + \cos \theta + \sin \theta)}$ है, तो $\frac{1 - \cos \theta + \sin \theta}{1 + \sin \theta}$ का मान है:

- a) $\frac{x}{(1+x)}$ b) x
c) $\frac{1}{x}$ d) $\frac{(1+x)}{x}$



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92. The value of $(2\cos^2\theta - 1) \left[\frac{1+\tan\theta}{1-\tan\theta} + \frac{1-\tan\theta}{1+\tan\theta} \right]$ is:
 $(2\cos^2\theta - 1) \left[\frac{1+\tan\theta}{1-\tan\theta} + \frac{1-\tan\theta}{1+\tan\theta} \right]$ का मान क्या है?

- a) 2 b) 0 c) $\frac{\sqrt{3}}{2}$ d) 1

93. If $x \sin^3\theta + y\cos^3\theta = \sin\theta \cos\theta$ and $x \sin\theta = y \cos\theta$, then the value of $x^2 + y^2$ is:
 यदि $x \sin^3\theta + y\cos^3\theta = \sin\theta \cos\theta$ और $x \sin\theta = y \cos\theta$, है तो $x^2 + y^2$ का मान है:

- a) 0 b) 4
c) 1 d) 2

94. If $\frac{\cos\alpha}{\sin\alpha+\cos\beta} + \frac{\cos\beta}{\sin\beta-\cos\alpha} = \frac{x}{\sin\alpha-\cos\beta} + \frac{\cos\beta}{\sin\beta+\cos\alpha}$, then x is equal to:
 यदि $\frac{\cos\alpha}{\sin\alpha+\cos\beta} + \frac{\cos\beta}{\sin\beta-\cos\alpha} = \frac{x}{\sin\alpha-\cos\beta} + \frac{\cos\beta}{\sin\beta+\cos\alpha}$, है तो x बराबर है:
 a) $\cos\beta$ b) $\cos\alpha$
c) $\sin\beta$ d) $\sin\alpha$

95. If $5\sin^2\theta - 4\cos\theta - 4 = 0$, $0^\circ < \theta < 90^\circ$, then the value of $(\cot\theta + \operatorname{cosec}\theta)$ is:
 यदि $5\sin^2\theta - 4\cos\theta - 4 = 0$, है, $0^\circ < \theta < 90^\circ$ है, तो $(\cot\theta + \operatorname{cosec}\theta)$ का मान ज्ञात करें।
 a) $\frac{3}{2}$ b) $\frac{\sqrt{6}}{2}$ c) $\frac{\sqrt{6}}{3}$ d) $\frac{2}{3}$

96. If $11\sin^2\theta - \cos^2\theta + 4\sin\theta - 4 = 0$, $0^\circ < \theta < 90^\circ$, then what is the value of $\frac{\cos 2\theta + \cot 2\theta}{\sec 2\theta - \tan 2\theta}$?
 यदि $11\sin^2\theta - \cos^2\theta + 4\sin\theta - 4 = 0$, $0^\circ < \theta < 90^\circ$ है, तो $\frac{\cos 2\theta + \cot 2\theta}{\sec 2\theta - \tan 2\theta}$ का मान क्या है?
 a) $\frac{10+5\sqrt{3}}{3}$ b) $\frac{12+7\sqrt{3}}{6}$
c) $\frac{10+7\sqrt{3}}{6}$ d) $\frac{12+5\sqrt{3}}{3}$

97. Solve the following equation.

$$\theta: 2\cos^2\theta + (4 + \sqrt{3})\sin\theta - 2(1 + \sqrt{3}) = 0 \text{ where } \theta \text{ is an acute angle.}$$

निम्नलिखित समीकरण को हल करें।

$$\theta: 2\cos^2\theta + (4 + \sqrt{3})\sin\theta - 2(1 + \sqrt{3}) = 0 \text{ जहाँ } \theta \text{ एक न्यून कोण है।}$$

- a) 30° b) 45°
c) 15° d) 60°

98. If $\frac{\sin^2\theta}{\cos^2\theta - 3\cos\theta + 2} = 1$, θ lies in the first quadrant, then the value of $\frac{\tan^2\frac{\theta}{2} + \sin^2\frac{\theta}{2}}{\tan\theta + \sin\theta}$ is:

यदि $\frac{\sin^2\theta}{\cos^2\theta - 3\cos\theta + 2} = 1$, θ प्रथम चतुर्थांश में स्थित है, तो $\frac{\tan^2\frac{\theta}{2} + \sin^2\frac{\theta}{2}}{\tan\theta + \sin\theta}$ का मान बताइए।
 a) $\frac{2\sqrt{3}}{27}$ b) $\frac{5\sqrt{3}}{27}$ c) $\frac{2\sqrt{3}}{9}$ d) $\frac{7\sqrt{3}}{54}$

99. For $\theta: 0^\circ < \theta < 90^\circ$ $3\sec\theta + 4\cos\theta = 4\sqrt{3}$, find the value of $(1 - \sin\theta + \cos\theta)$.
 θ के लिए: $0^\circ < \theta < 90^\circ$ है। $3\sec\theta + 4\cos\theta = 4\sqrt{3}$ है, तो $(1 - \sin\theta + \cos\theta)$ का मान ज्ञात करें।

- a) $\frac{1+2\sqrt{3}}{2}$ b) $\frac{1+\sqrt{3}}{2}$ c) $\frac{1-\sqrt{3}}{2}$ d) $\frac{1-2\sqrt{3}}{2}$

100. If $\sec\theta - 2\cos\theta = \frac{7}{2}$, where θ is a positive acute angle, then the value of $\sec\theta$ is:
 यदि $\sec\theta - 2\cos\theta = \frac{7}{2}$ है, जहाँ θ एक धनात्मक न्यून कोण है, तो $\sec\theta$ का मान ज्ञात कीजिए।

- a) 6 b) 8
c) 5 d) 4

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101. If $3\sin\theta = 2\cos^2\theta$, $0^\circ < \theta < 90^\circ$, then the value of $(\tan^2\theta + \sec^2\theta - \operatorname{cosec}^2\theta)$ is:

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यदि $3\sin\theta = 2\cos^2\theta$, $0^\circ < \theta < 90^\circ$, तो $(\tan^2\theta + \sec^2\theta - \operatorname{cosec}^2\theta)$ का मान है:

- a) -2 b) $-\frac{7}{3}$
c) $\frac{7}{3}$ d) 2

102. If $\sin\theta \cdot \sec^2\theta = \frac{2}{3}$, $0^\circ < \theta < 90^\circ$, then the value of $(\tan^2\theta + \cos^2\theta)$ is:

यदि $\sin\theta \cdot \sec^2\theta = \frac{2}{3}$, $0^\circ < \theta < 90^\circ$ है, तो $(\tan^2\theta + \cos^2\theta)$ का मान है:

- a) $\frac{7}{6}$ b) $\frac{11}{12}$
c) $\frac{13}{12}$ d) $\frac{5}{4}$

103. If $1 + \sin^2\theta - 3\sin\theta \cos\theta = 0$, then the value of $\cot\theta$ is:

यदि $1 + \sin^2\theta - 3\sin\theta \cos\theta = 0$, तो $\cot\theta$ का मान क्या होगा?

- a) 0 b) 2 c) $\frac{1}{2}$ d) $\frac{1}{3}$

104. If $6\tan\theta - 5\sqrt{3}\sec\theta + 12\cot\theta = 0$, $0^\circ < \theta < 90^\circ$, then the value of $(\operatorname{cosec}\theta + \sec\theta)$ is:

यदि $6\tan\theta - 5\sqrt{3}\sec\theta + 12\cot\theta = 0$, $0^\circ < \theta < 90^\circ$, तो $(\operatorname{cosec}\theta + \sec\theta)$ का मान है:

- a) $\frac{3+2\sqrt{3}}{2}$ b) $\frac{2}{3}(3 + \sqrt{3})$
c) $\frac{3+\sqrt{3}}{2}$ d) $\frac{2(3+2\sqrt{3})}{3}$

105. If $\sec\theta$ and $\sin\theta$ ($0 < \theta < 90$) are the roots of the equation $\sqrt{6}x^2 - kx + \sqrt{6} = 0$, then the value of k is:

यदि $\sec\theta$ और $\sin\theta$ ($0 < \theta < 90$) समीकरण $\sqrt{6}x^2 - kx + \sqrt{6} = 0$, के मूल हैं, तो k का मान है:

- a) $\sqrt{3}$ b) $3\sqrt{2}$
c) $2\sqrt{3}$ d) $3\sqrt{3}$

106. If $\sin(A+B) = \cos(A+B)$, what is the value of $\tan A$?

यदि $\sin(A+B) = \cos(A+B)$ है, तो $\tan A$ का मान कितना होगा?

- a) $\frac{1-\tan B}{1+\tan B}$ b) $\frac{1+\tan B}{1-\tan B}$
c) $\frac{1+\sec B}{1-\sec B}$ d) $\frac{1-\operatorname{cosec} B}{1+\operatorname{cosec} B}$

107. Simplify the following:

$$\frac{\cos x - \sqrt{3}\sin x}{2}$$

निम्नलिखित को सरल करें:

$$\frac{\cos x - \sqrt{3}\sin x}{2}$$

- a) $\cos\left(\frac{\pi}{3} - x\right)$ b) $\sin\left(\frac{\pi}{3} + x\right)$
c) $\cos\left(\frac{\pi}{3} + x\right)$ d) $\sin\left(\frac{\pi}{3} - x\right)$

108. The value of θ when $\sqrt{3}\cos\theta + \sin\theta = 1$, ($0^\circ \leq \theta \leq 90^\circ$), is:

θ का मान क्या होगा, जब $\sqrt{3}\cos\theta + \sin\theta = 1$, ($0^\circ \leq \theta \leq 90^\circ$) है?

- a) 90° b) 30°
c) 60° d) 0°

109. If $\tan A \tan B + \frac{\cos x}{\cos A \cos B} = 1$, then $x = ?$

यदि $\tan A \tan B + \frac{\cos x}{\cos A \cos B} = 1$ है, तो $x = ?$

- a) B b) A
c) $A + B$ d) $A - B$

110. If $\tan(\alpha + \beta) = a$, $\tan(\alpha - \beta) = b$, then the value of $\tan 2\alpha$ is:

यदि $\tan(\alpha + \beta) = a$, $\tan(\alpha - \beta) = b$ है, तो $\tan 2\alpha$ का मान ज्ञात कीजिए।

- a) $\frac{a+b}{1-ab}$ b) $\frac{a+b}{1+ab}$
c) $\frac{a-b}{1+ab}$ d) $\frac{a-b}{1-ab}$

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121. If $\cos^2\theta + 3 = 3(\cot^2\theta + \sin^2\theta)$, $0^\circ < \theta < 90^\circ$, then what is the value of $(2\cos\theta + 3\sin\theta)$?

यदि $\cos^2\theta + 3 = 3(\cot^2\theta + \sin^2\theta)$, $0^\circ < \theta < 90^\circ$ है, तो $(2\cos\theta + 3\sin\theta)$ का मान कितना होगा?

- a) $\frac{1+3\sqrt{3}}{2}$ b) $\frac{2\sqrt{3}+1}{2}$
c) $\frac{\sqrt{3}+2}{2}$ d) $\frac{2+3\sqrt{3}}{2}$

122. If $\cos A, \sin A, \cot A$ are in geometric progression, then the value of $\tan^6 A - \tan^2 A$ is:

यदि $\cos A, \sin A, \cot A$ गुणोत्तर श्रेणी में हैं, तो $\tan^6 A - \tan^2 A$ का मान है :

- a) $1/2$ b) 3
c) $1/3$ d) 1

123. If $\cos^2\theta - \sin^2\theta = \tan^2\phi$, then which of the following is true?

यदि $\cos^2\theta - \sin^2\theta = \tan^2\phi$ है तो निम्न में से कौन सा सत्य है?

- a) $\cos\theta \cos\phi = 1$
b) $\cos\theta \cos\phi = \sqrt{2}$
c) $\cos^2\phi - \sin^2\phi = \cot^2\theta$
d) $\cos^2\phi - \sin^2\phi = \tan^2\theta$

124. If $\frac{\sec\theta - \tan\theta}{\sec\theta + \tan\theta} = \frac{1}{7}$, θ lies in first quadrant, then the value of $\frac{\csc\theta + \cot^2\theta}{\csc\theta - \cot^2\theta}$ is:

यदि $\frac{\sec\theta - \tan\theta}{\sec\theta + \tan\theta} = \frac{1}{7}$, θ प्रथम चतुर्थांश में स्थित है, तो $\frac{\csc\theta + \cot^2\theta}{\csc\theta - \cot^2\theta}$ का मान बताइए।

- a) $\frac{19}{5}$ b) $\frac{22}{3}$ c) $\frac{37}{12}$ d) $\frac{37}{19}$

125. If $\sec\theta - \cos\theta = 14$ and $14\sec\theta = x$, then the value of x is _____.

यदि $\sec\theta - \cos\theta = 14$ and $14\sec\theta = x$ है, तो x का मान ज्ञात कीजिए।

- a) $\sec^2\theta$ b) $\tan^2\theta$
c) $2\tan\theta$ d) $2\sec\theta$

126. The least value of $8\csc^2\theta + 25\sin^2\theta$ is:

$8\csc^2\theta + 25\sin^2\theta$ का न्यूनतम मान है:

- a) $20\sqrt{2}$ b) $10\sqrt{2}$
c) $40\sqrt{2}$ d) $30\sqrt{2}$

127. If $A + B = 45^\circ$, then the value of $2(1 + \tan A)(1 + \tan B)$ is:

यदि $A + B = 45^\circ$, तो $2(1 + \tan A)(1 + \tan B)$ का मान है:

- a) 2 b) 4
c) 1 d) 0

128. If $1 + \sin\theta = m\cos\theta$, then what is the value of $\sin\theta$?

यदि $1 + \sin\theta = m\cos\theta$ है, तो $\sin\theta$ का मान क्या होगा?

- a) $\frac{2m^2-1}{m^2+1}$ b) $\frac{m^2-1}{m^2+1}$
c) $\frac{m^2+1}{2m^2-1}$ d) $\frac{m^2+1}{m^2-1}$

129. Which of the following is the value of

$$\sqrt{\frac{1-\sin 45^\circ}{1+\sin 45^\circ}}?$$

$\sqrt{\frac{1-\sin 45^\circ}{1+\sin 45^\circ}}$ का मान निम्नलिखित में से कौन-सा है?

- a) $\cos 45^\circ - \tan 45^\circ$
b) $\tan 45^\circ - \sec 45^\circ$
c) $\tan 45^\circ$
d) $\sec 45^\circ - \tan 45^\circ$

130. If $\frac{x - x\csc^2 30^\circ}{1 + \csc^2 30^\circ} = \cos^2 60^\circ + 4\cot^2 45^\circ - \sec^2 60^\circ$, then the value of x is _____.

यदि $\frac{x - x\csc^2 30^\circ}{1 + \csc^2 30^\circ} = \cos^2 60^\circ + 4\cot^2 45^\circ - \sec^2 60^\circ$ है, तो x का मान ज्ञात कीजिए।

- a) $-\frac{1}{12}$ b) $\frac{5}{12}$
c) $-\frac{5}{12}$ d) $\frac{1}{5}$

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131. If $\frac{5\cot\theta + \sqrt{3}\operatorname{cosec}\theta}{2\sqrt{3}\operatorname{cosec}\theta + 3\cot\theta} = 1$, $\theta < \theta < 90^\circ$, then the

value of $\frac{\frac{7}{2}\cot^2\theta - \frac{3}{4}\operatorname{cosec}^2\theta}{4\sin^2\theta + \frac{3}{2}\tan^2\theta}$ will be:

यदि $\frac{5\cot\theta + \sqrt{3}\operatorname{cosec}\theta}{2\sqrt{3}\operatorname{cosec}\theta + 3\cot\theta} = 1$, $\theta < \theta < 90^\circ$ है, तो

$\frac{\frac{7}{2}\cot^2\theta - \frac{3}{4}\operatorname{cosec}^2\theta}{4\sin^2\theta + \frac{3}{2}\tan^2\theta}$ का मान ज्ञात करें।

a) 7 b) 2 c) 3 d) 5

132. For all α_i 's, ($i = 1, 2, 3, \dots, 20$) lying from 0° to 90° , it is given that, $\sin \alpha_1 + \sin \alpha_2 + \sin \alpha_3 + \dots + \sin \alpha_{20} = 20$. What is the value of $(\alpha_1 + \alpha_2 + \alpha_3 + \dots + \alpha_{20})$?

सभी α_i ($i = 1, 2, 3, \dots, 20$) के लिए, जो कि 0° से 90° तक हैं, यह दिया गया है कि $\sin \alpha_1 + \sin \alpha_2 + \sin \alpha_3 + \dots + \sin \alpha_{20} = 20$ है। तो $(\alpha_1 + \alpha_2 + \alpha_3 + \dots + \alpha_{20})$ का मान क्या होगा?

a) 1800° b) 900° c) 0° d) 20°

133. For all α_i 's, ($i = 1, 2, 3, \dots, 20$) lying from 0° to 90° , it is given that, $\cos \alpha_1 + \cos \alpha_2 + \cos \alpha_3 + \dots + \cos \alpha_{20} = 20$. What is the value of $(\alpha_1 + \alpha_2 + \alpha_3 + \dots + \alpha_{20})$?

सभी α_i , ($i = 1, 2, 3, \dots, 20$) के लिए, जो कि 0° से 90° तक हैं, यह दिया गया है कि $\cos \alpha_1 + \cos \alpha_2 + \cos \alpha_3 + \dots + \cos \alpha_{20} = 20$ है। तो $(\alpha_1 + \alpha_2 + \alpha_3 + \dots + \alpha_{20})$ का मान क्या होगा?

a) 900° b) 1800°
c) 0° d) 20°

134. If $\cos A + \cos B + \cos C = 3$, then what is the value of $\sin A + \sin B + \sin C$?

यदि $\cos A + \cos B + \cos C = 3$, तो $\sin A + \sin B + \sin C$ का मान क्या है?

a) 1 b) 2 c) 0 d) -1

135. If $\sin \alpha + \sin \beta = \cos \alpha + \cos \beta = 1$, then $\sin \alpha + \cos \alpha = ?$

यदि $\sin \alpha + \sin \beta = \cos \alpha + \cos \beta = 1$ है, तो $\sin \alpha + \cos \alpha = ?$

a) 2 b) 1 c) 0 d) -1

136. Using $2\cos A \cos B = \cos(A+B) + \cos(A-B)$, find the value of $\cos 75^\circ \cos 15^\circ$.

$2\cos A \cos B = \cos(A+B) + \cos(A-B)$ के उपयोग से $\cos 75^\circ \cos 15^\circ$ का मान ज्ञात कीजिए।

a) $\frac{1}{2}$ b) $\frac{\sqrt{2}}{4}$ c) $\frac{\sqrt{2}}{2}$ d) $\frac{1}{4}$

137. If $\sin A = \frac{5}{12}$ and $7 \cot B = 24$, then the value of $(\sec A \cos B)(\operatorname{cosec} B \tan A)$ is:

यदि $\sin A = \frac{5}{12}$ और $7 \cot B = 24$ है, तो $(\sec A \cos B)(\operatorname{cosec} B \tan A)$ का मान क्या होगा?

a) $\frac{13}{14}$ b) $\frac{15}{13}$ c) $\frac{13}{7}$ d) $\frac{65}{42}$

138. If $b \cos \theta = a$, then $\operatorname{cosec} \theta + \cot \theta = \underline{\hspace{2cm}}$.

यदि $b \cos \theta = a$ है, तो $\operatorname{cosec} \theta + \cot \theta = \underline{\hspace{2cm}}$ होगा।

a) $\sqrt{\frac{1}{b+a}}$ b) $\sqrt{\frac{b-a}{b+a}}$
c) $\sqrt{\frac{b+a}{b-a}}$ d) $\sqrt{\frac{1}{b-a}}$

139. In the given figure, ABC is an isosceles triangle with $BC = 8\text{cm}$ and $AB = AC = 5\text{cm}$. The value of $\tan C - \cot B$ is $\underline{\hspace{2cm}}$.

दी गई आकृति में, ABC एक समद्विबाहु त्रिभुज है जिसमें $BC = 8\text{cm}$ और $AB = AC = 5\text{cm}$ है। $\tan C - \cot B$ का मान $\underline{\hspace{2cm}}$ है।



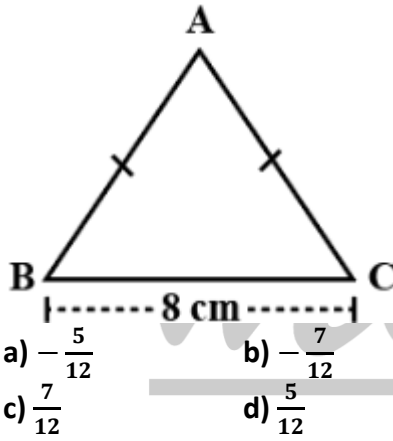
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एक क्लॉक टावर उत्तर-दक्षिण और पूर्व-पश्चिम दिशाओं में किस बिंदु की ओर चौराहे पर खड़ा है। P, Q, R और S क्रमशः उत्तर, पूर्व, दक्षिण और पश्चिम की ओर जाने वाली सड़कों पर बिंदु हैं, जहाँ टावर के शीर्ष के उन्नयन कोण क्रमशः α, β, γ and δ हैं। तो $\left(\frac{PQ}{RS}\right)^2$ के बराबर है।

- a) $\frac{\tan^2 \alpha + \tan^2 \delta}{\tan^2 \gamma + \tan^2 \beta}$ b) $\frac{\tan^2 \alpha + \tan^2 \beta}{\tan^2 \gamma + \tan^2 \delta}$
 c) $\frac{\cot^2 \alpha + \cot^2 \beta}{\cot^2 \gamma + \cot^2 \delta}$ d) $\frac{\cot^2 \alpha + \cot^2 \delta}{\cot^2 \gamma + \cot^2 \beta}$

ANSWER KEY

1. D	2. A	3. D	4. A	5. C
6. C	7. C	8. A	9. D	10. B
11. B	12. B	13. D	14. D	15. B
16. A	17. D	18. C	19. A	20. B
21. A	22. D	23. B	24. A	25. A
26. C	27. B	28. A	29. C	30. D
31. B	32. C	33. D	34. B	35. A
36. A	37. D	38. D	39. A	40. D
41. D	42. B	43. C	44. B	45. A
46. C	47. D	48. C	49. A	50. C
51. A	52. B	53. C	54. A	55. B
56. D	57. A	58. A	59. A	60. C
61. C	62. B	63.	64. D	65. A
66. D	67. D	68. C	69. B	70. D
71. D	72. C	73. B	74. C	75. D
76. A	77. C	78. B	79. C	80. D
81. C	82. C	83. A	84. A	85. B
86. D	87. D	88. B	89. A	90. C
91. C	92. A	93. C	94. B	95. B
96. B	97. D	98. D	99. B	100. D
101. B	102. C	103. B	104. B	105. D
106. A	107. C	108. A	109. C	110. A
111. C	112. C	113. A	114. B	115. C
116. A	117. D	118. C	119. A	120. C
121. B	122. D	123. D	124. A	125. B
126. A	127. B	128. B	129. D	130. C
131. D	132. A	133. B	134. C	135. B
136. D	137. D	138. C	139. B	140. C
141. B	142. C			

140. If $\cos \theta = \frac{4x}{1+4x^2}$ then what is the value of $\sin \theta$?

यदि $\cos \theta = \frac{4x}{1+4x^2}$ तो $\sin \theta$ का मान क्या होगा?

- a) $\frac{1+4x^2}{1-4x^2}$ b) $\frac{1+4x^2}{4x^2}$
 c) $\frac{1-4x^2}{1+4x^2}$ d) $\frac{1-4x^2}{4x^2}$

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141. If $\sin \theta = \frac{2\sqrt{ab}}{a+b}$, $a > b > 0$, then the value of $\frac{\cos \theta + 1}{\cos \theta - 1}$ will be:

यदि $\sin \theta = \frac{2\sqrt{ab}}{a+b}$ है, $a > b > 0$ है, तो $\frac{\cos \theta + 1}{\cos \theta - 1}$ का मान ज्ञात करें।

- a) $-\frac{b}{a}$ b) $-\frac{a}{b}$ c) $\frac{a}{b}$ d) $\frac{b}{a}$

142. A clock tower stands at the crossing of towards which point in the north-south and the east – west directions. P, Q, R and S are points on the roads due north, east, south and west respectively, where the angles of elevation of the top of the tower are respectively, α, β, γ and δ . Then $\left(\frac{PQ}{RS}\right)^2$ is equal to.



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